Roma, 18 Marzo 2016

{codemotion}



You Turing-Complete Me

L’arte di programmare (perdere ancora più tempo) dentro i videogiochi



Ciao!

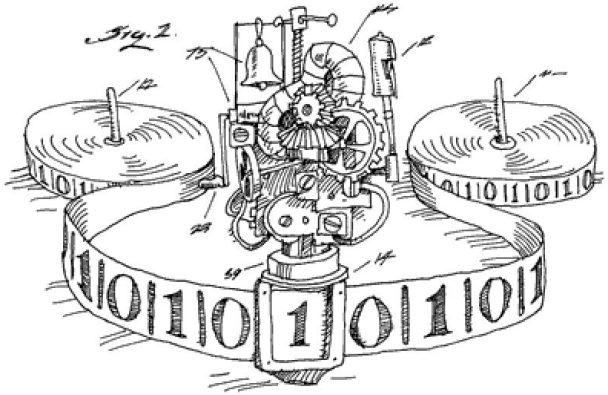
Siamo Sara e Luca Di Bartolomeo

Studiamo informatica, ci dilettiamo nella creazione
di piccoli videogiochi e insegnamo ai bambini di
Codemotion Kids a programmare usando Minecraft.

1.
Macchine di Turing

Cinque minuti di teoria prima della parte divertente
(vi prometto che durerà poco)

Cos'è una macchina di Turing



Una macchina di Turing è una macchina astratta che:

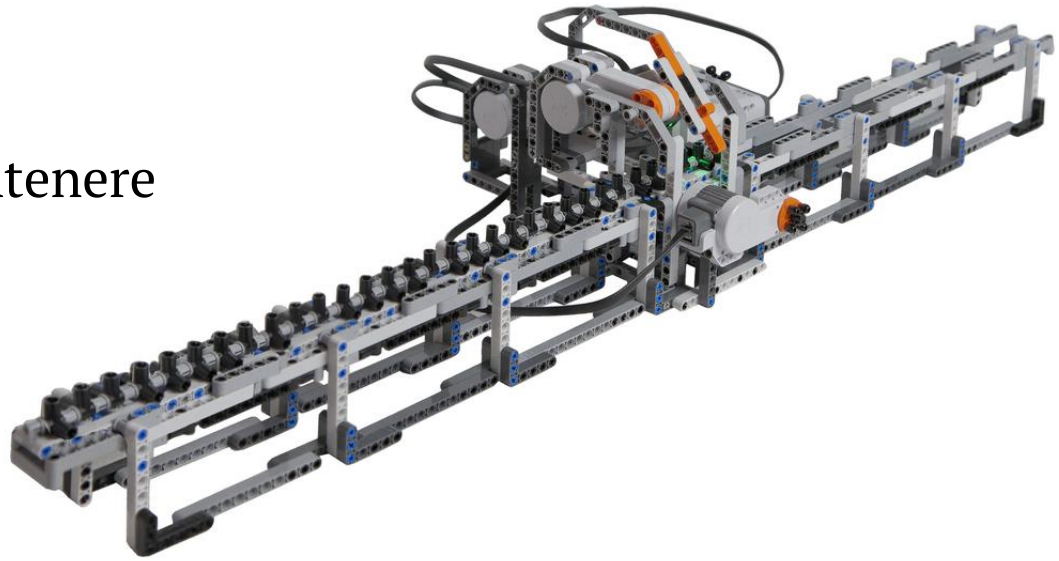
- Legge dei simboli in input
- Esegue delle azioni seguendo delle regole

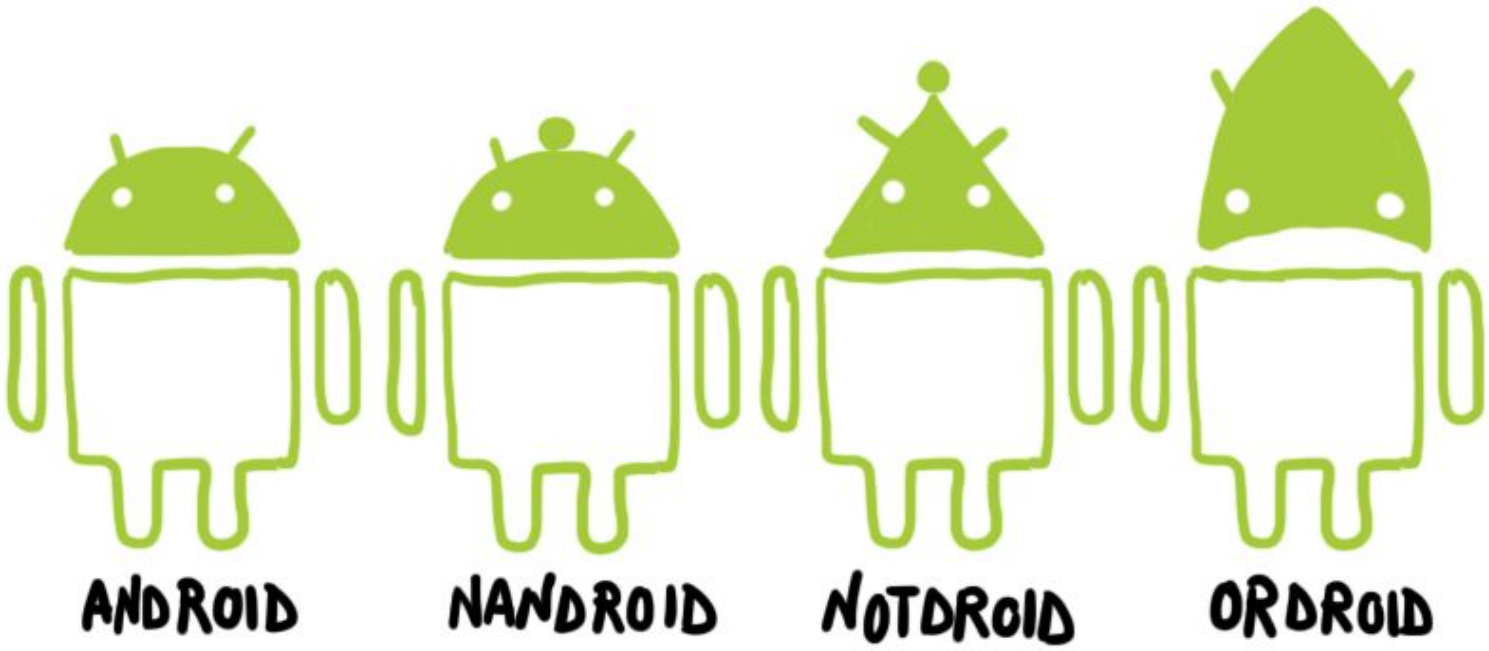
Una macchina di Turing universale è una macchina che può simulare tutte le altre macchine di Turing.

A partire da un meccanismo basilare è possibile simulare ogni possibile operazione che può essere eseguita da un computer.

Ingredienti:

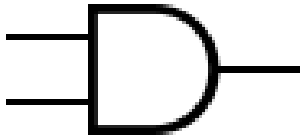
- Lettura di input
- Capacità di mantenere uno stato
- Svolgimento di operazioni basilari




ANDROID
NANDROID
NOTDROID
ORDROID

Le porte logiche sono i mattoncini basilari per la costruzione di un circuito digitale

AND



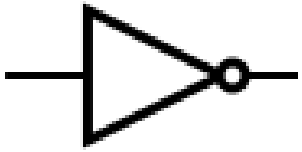
A	B	A and B
1	1	1
1	0	0
0	1	0
0	0	0

OR



A	B	A or B
1	1	1
1	0	1
0	1	1
0	0	0

NOT



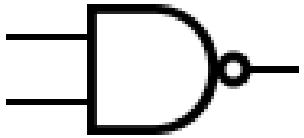
A	Not A
1	0
0	1

XOR

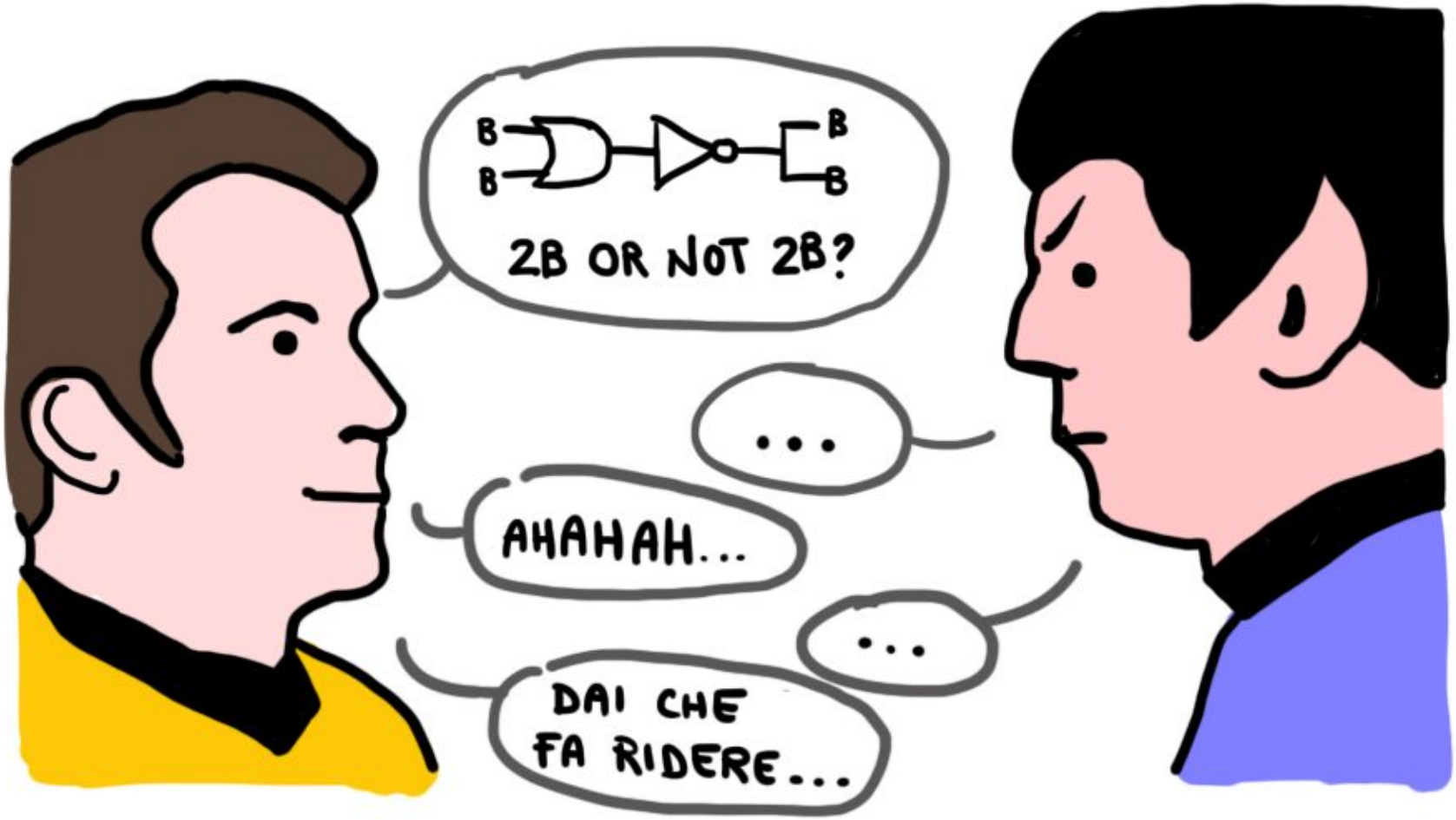


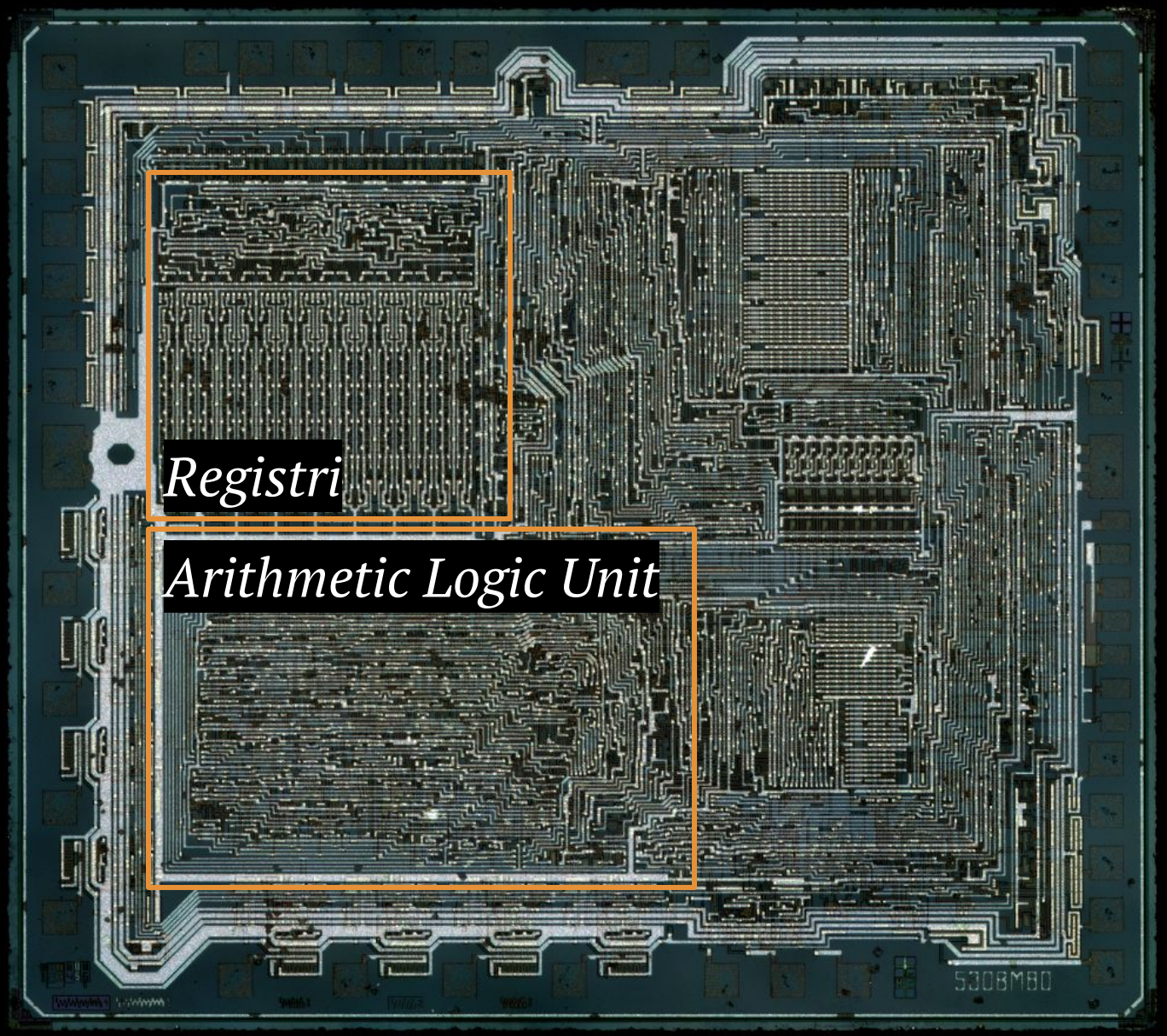
A	B	A xor B
1	1	0
1	0	1
0	1	1
0	0	0

NAND



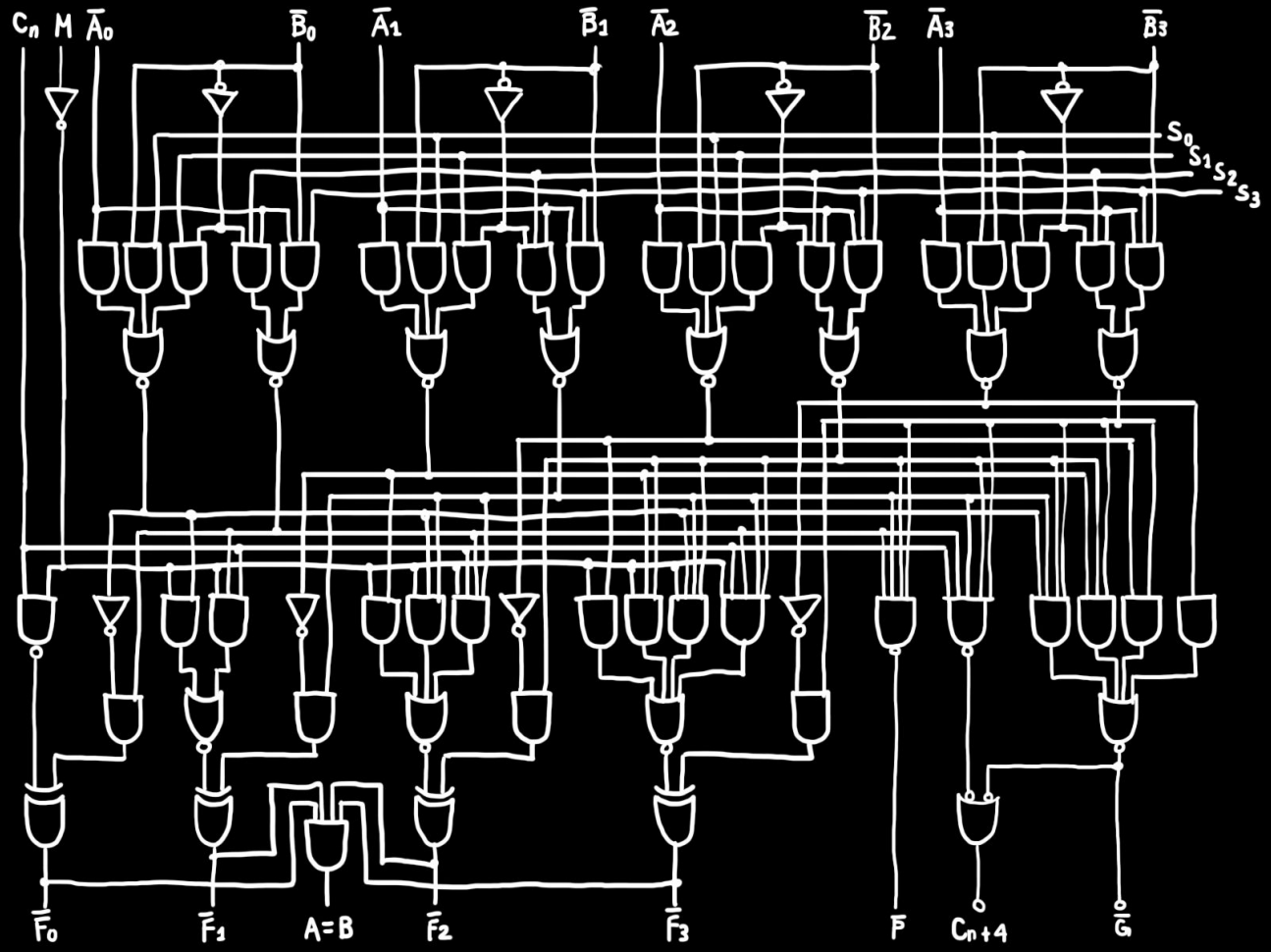
A	B	A nand B
1	1	0
1	0	1
0	1	1
0	0	1


B
B
B
B
2B OR NOT 2B?
...
AHAHAH...
...
DAI CHE
FA RIDERE...

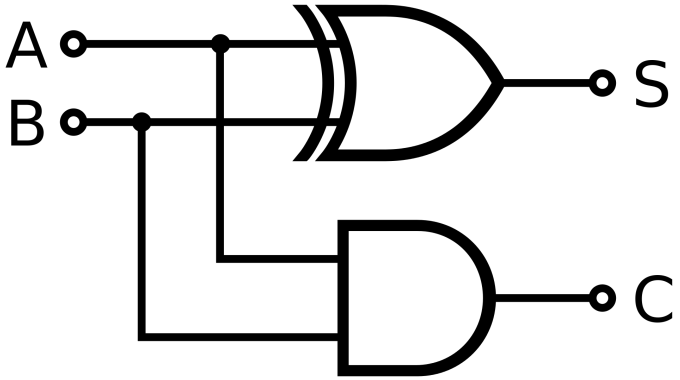

Registri
Arithmetic Logic Unit
5308M80


WE NEED TO GO DEEPER

Com’è fatta un’ALU:

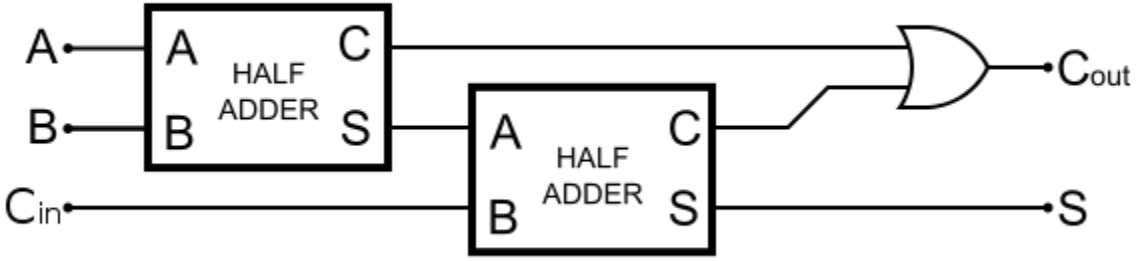


Half Adder

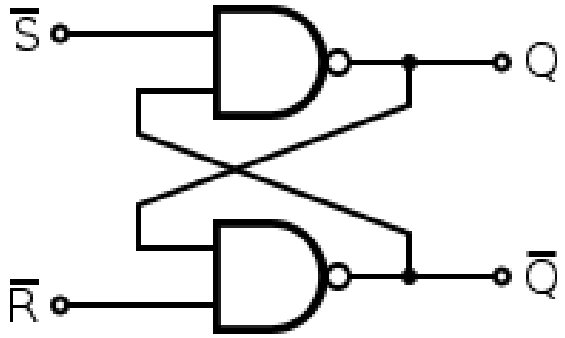


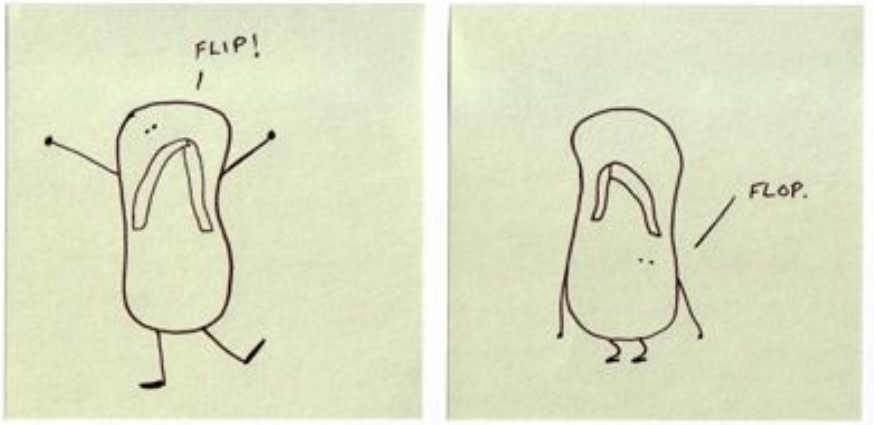
Input		Output	
A	B	S	C
1	1	0	1
1	0	1	0
0	1	1	0
0	0	0	0

Full Adder



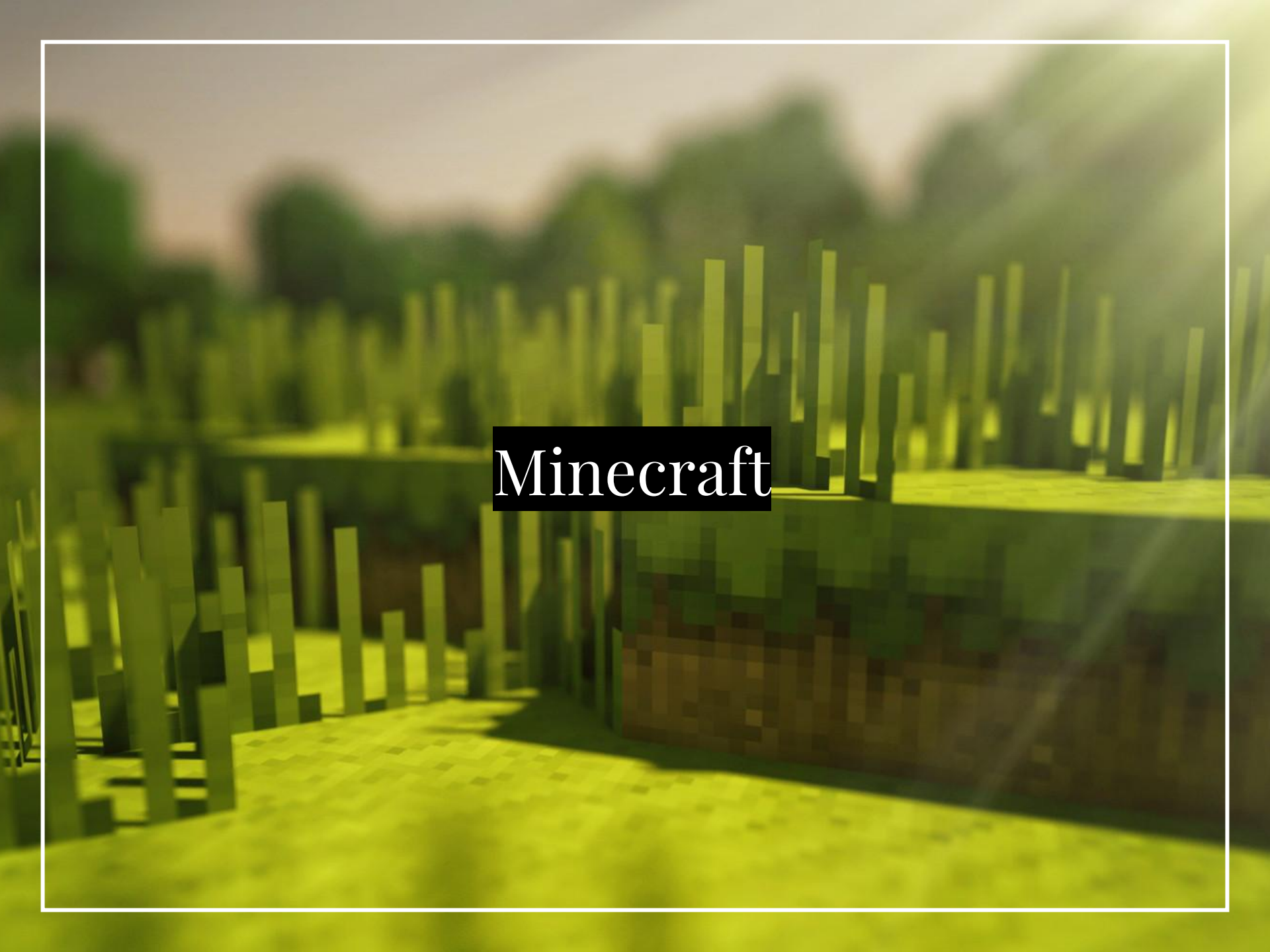
Flip Flop (SR latch)





E' sufficiente poter costruire un AND e un OR per poter dire che un sistema è turing-completo?

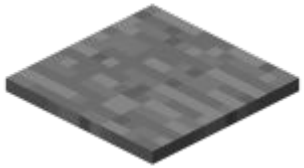
(La risposta è: sì)


Minecraft

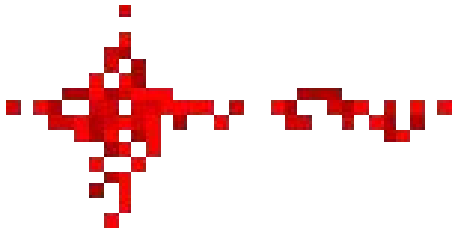
Redstone: simula il comportamento di un filo attraverso cui può passare un segnale

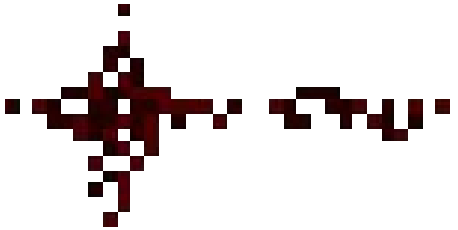
Input











Output







YOU MUST BE A REDSTONE TORCH, BECAUSE YOU ARE EXTENDING MY PISTON.

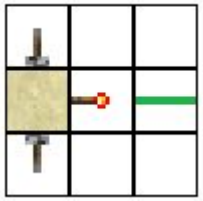




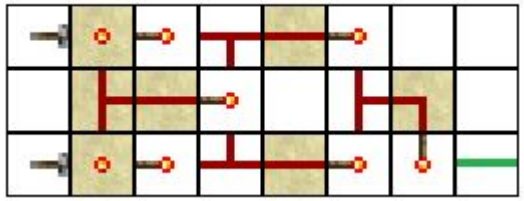
NOT Gate
(Inverter)



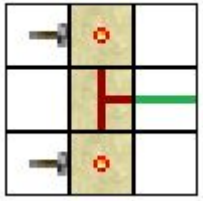
NOR Gate



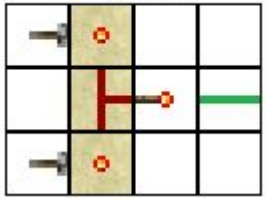
XNOR Gate



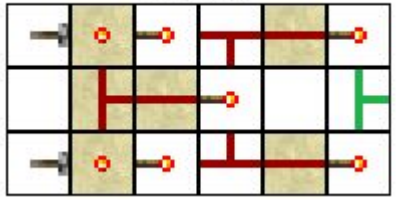
NAND Gate



AND Gate



XOR Gate



Si può ricostruire Minecraft dentro Minecraft?

256 mb

La quantità minima di RAM per eseguire Minecraft

2,147,483,648 flip flop

Per ricostruire 256 mb

53 * 10^9

Blocchi necessari, considerando flip flop grandi 5x5, per 256 mb

65536

Blocchi in un chunk, un chunk pesa 6 mb

819200

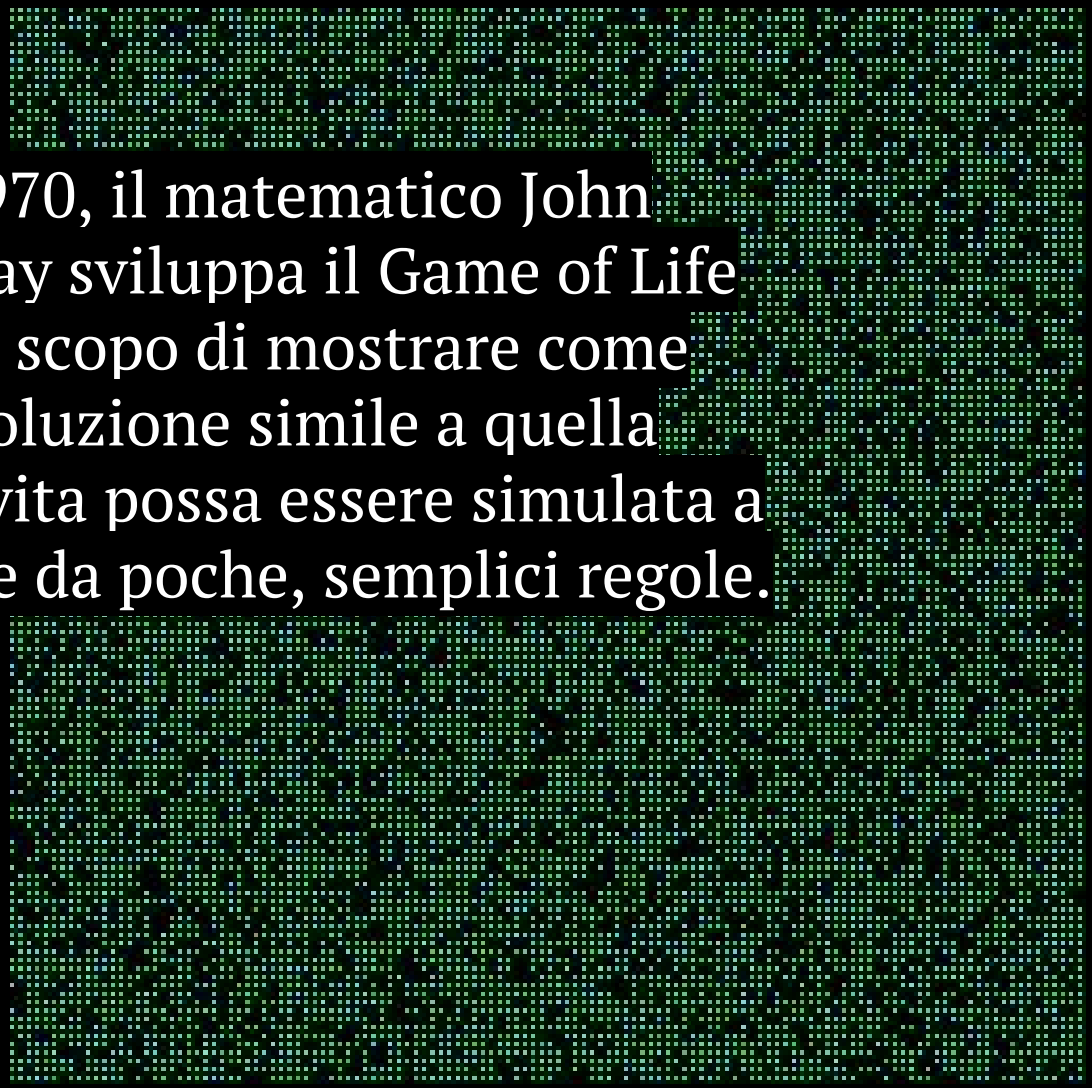
Per contenere 53 * 10^9 blocchi

4915200 mb = 4,7 tb

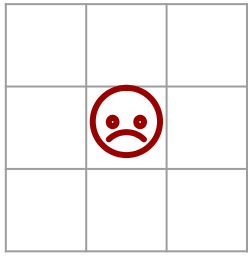
Memoria necessaria per contenere tutti quei chunk


Conway’s Game of Life

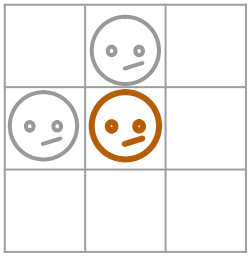
Nel 1970, il matematico John Conway sviluppa il Game of Life con lo scopo di mostrare come un’evoluzione simile a quella della vita possa essere simulata a partire da poche, semplici regole.



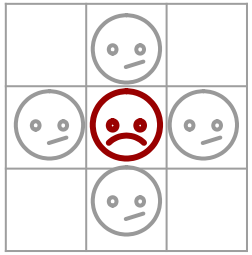
Il gioco si svolge su una griglia di dimensioni arbitrarie.
Ogni cella della griglia può essere viva o morta.



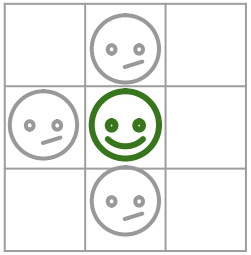
Se una cella viva non ha nessun vicino vivo, oppure un solo vicino vivo, muore di solitudine.



Se una cella viva ha 2 vicini vivi, non cambia il suo stato.



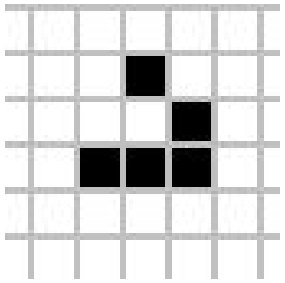
Se una cella viva ha 4 vicini vivi, muore per sovraffollamento.



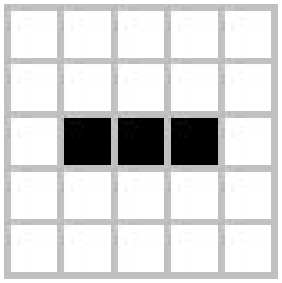
Se una cella ha tre vicini vivi, si riproduce.

Alcune configurazioni sono particolarmente interessanti:

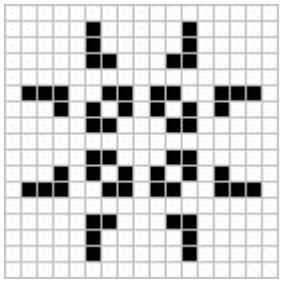
Glider:



Blinker:



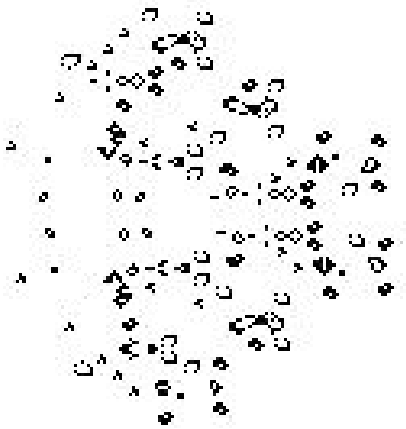
Pulsar:



Glider Gun:



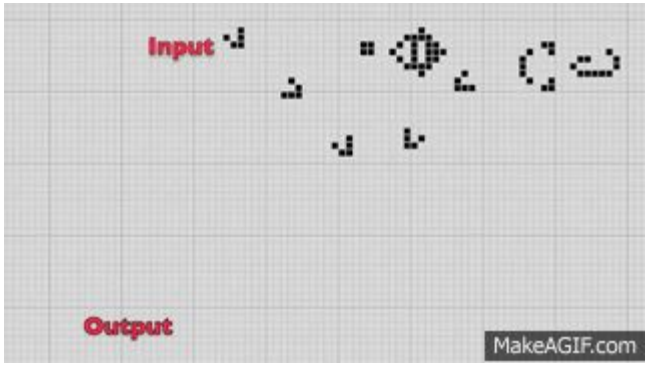
Un esempio:



Porte logiche nel game of life

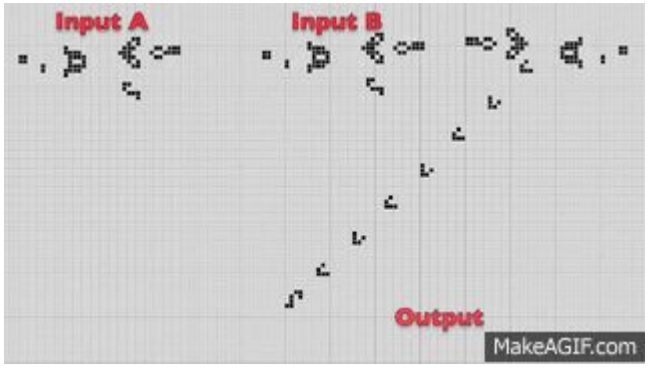
NOT



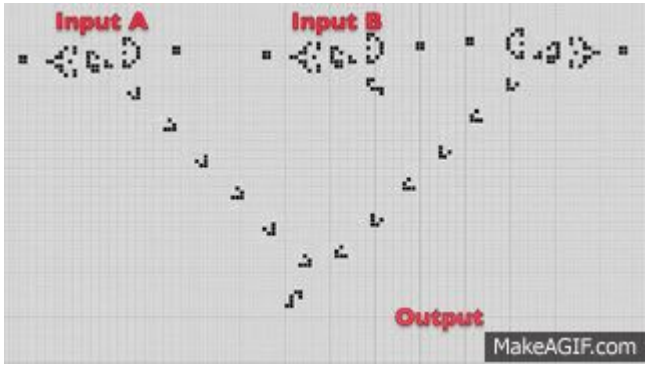


AND

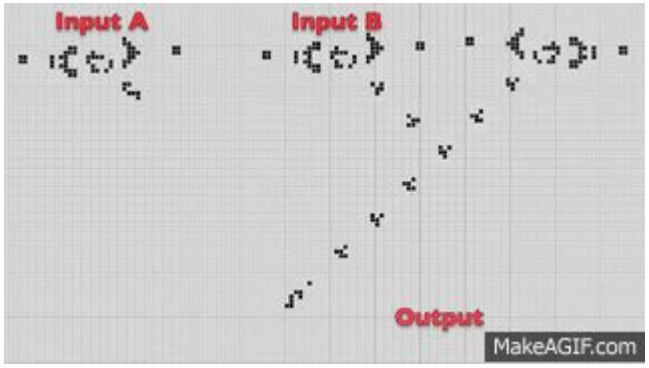
A = 0, B = 0, out = 0



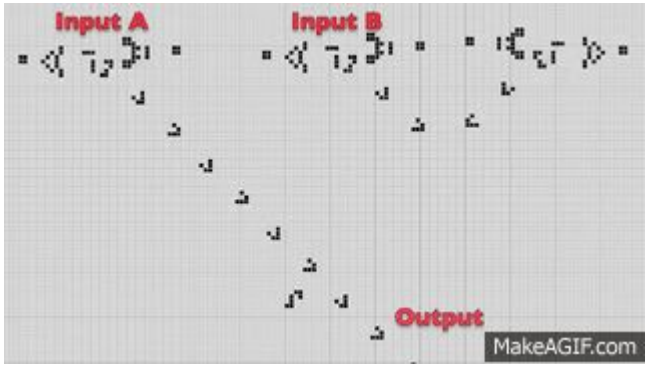
A = 1, B = 0, out = 0



A = 0, B = 1, out = 0

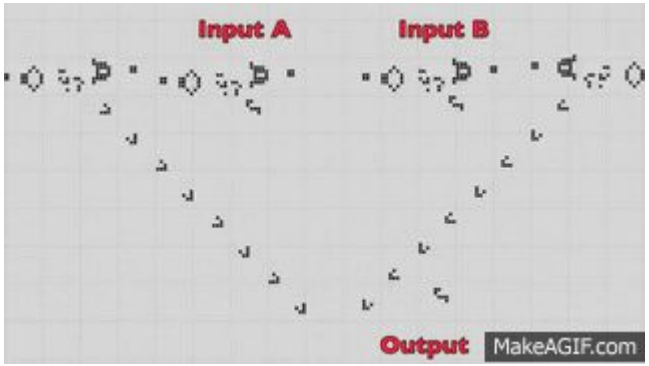


A = 1, B = 1, out = 1

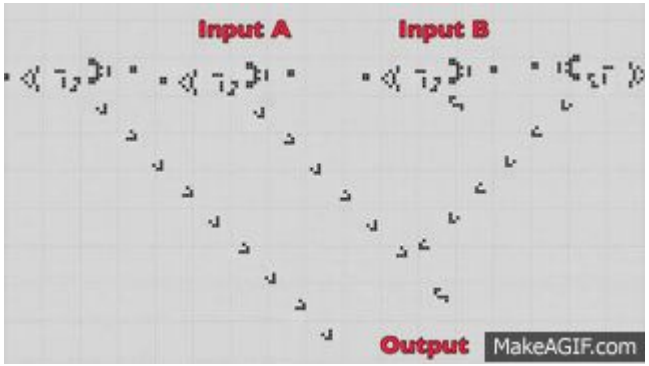


OR

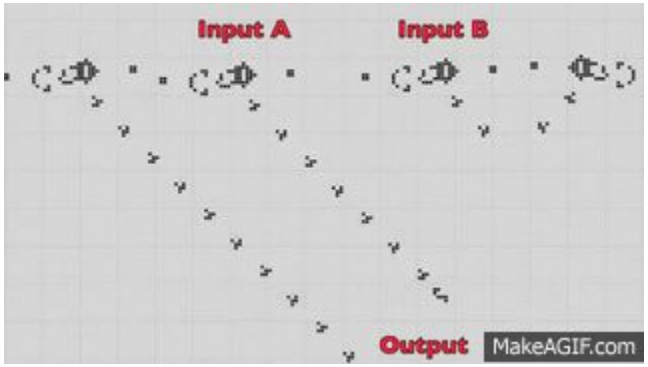
A = 0, B = 0, out = 0

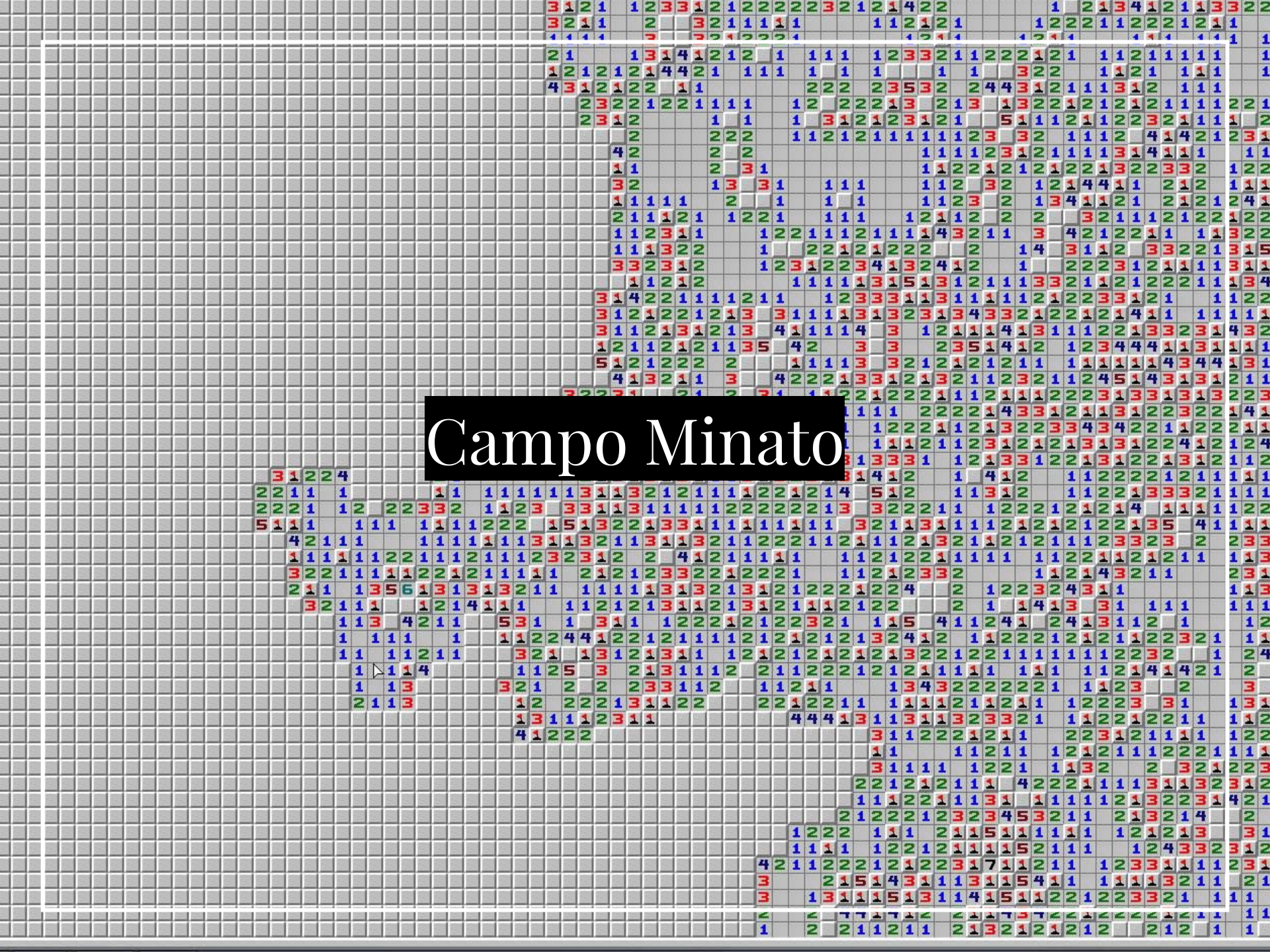


A = 1, B = 0, out = 1

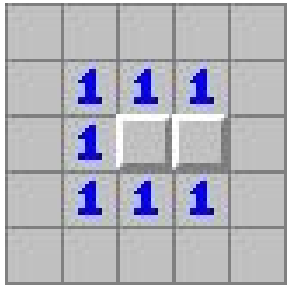


A = 1, B = 1, out = 1

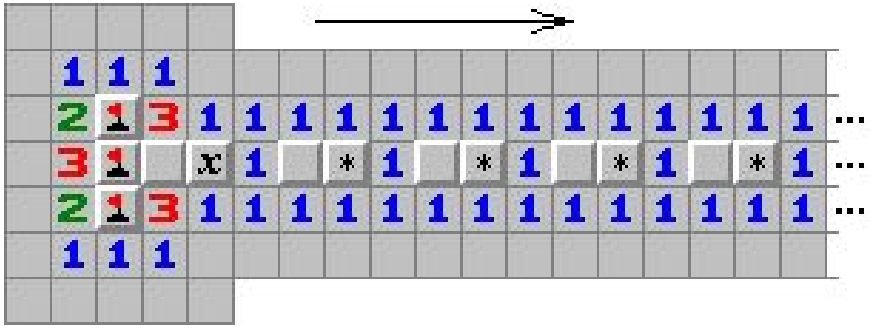



Campo Minato

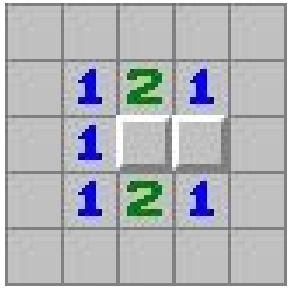
Consistente:



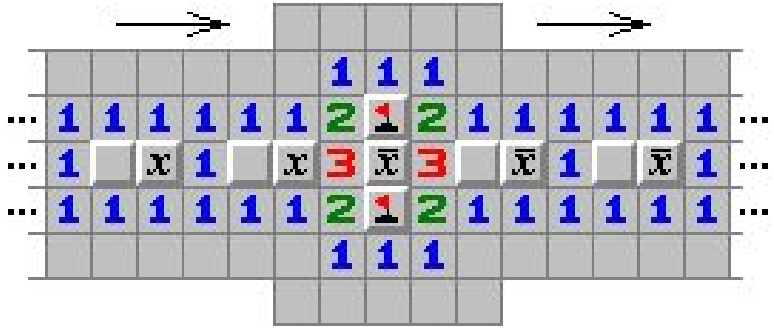
Trasportare un segnale:



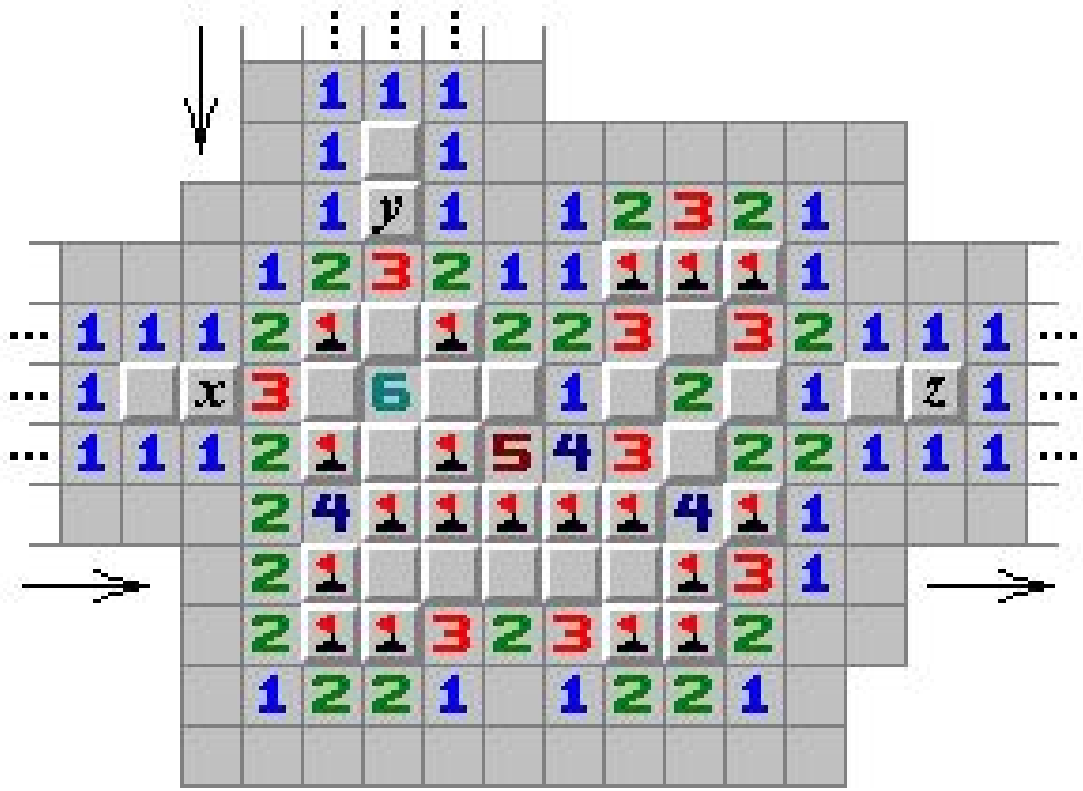
Inconsistente:



Negare un segnale:



Un Or in campo minato:





Pokemon Giallo

Basic Pokémon

g[illegible] HP



Pokémon. Length: A[illegible], Weight: b

Pokémon Power: Self Destruct Throw all the cards off the table and break the game.

Exploit Transform the game into another game.

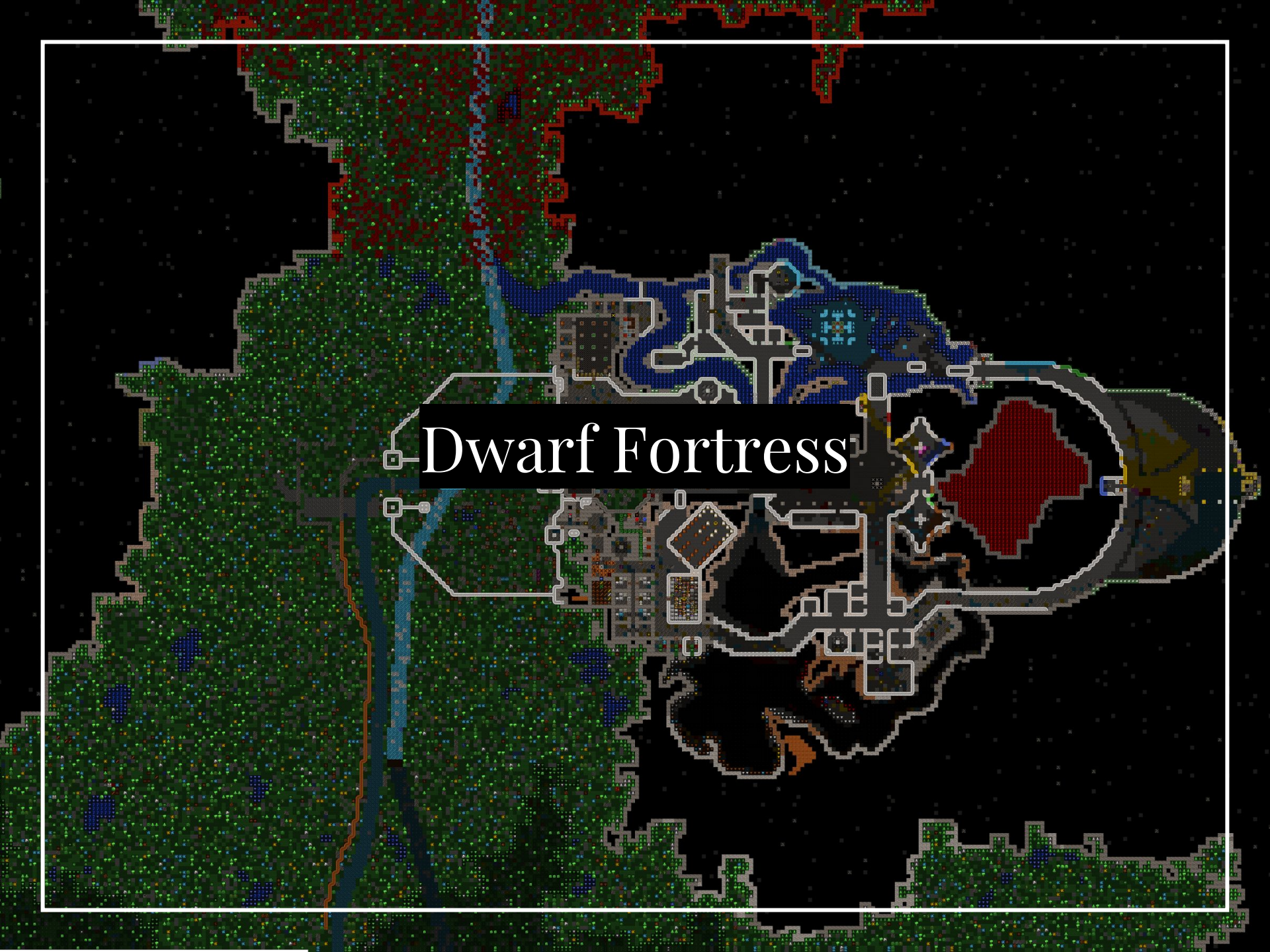
nan

weakness	resistance	retreat cost

i[illegible]c[illegible]n[illegible]id[illegible]h[illegible]B[illegible]V[illegible]l[illegible]

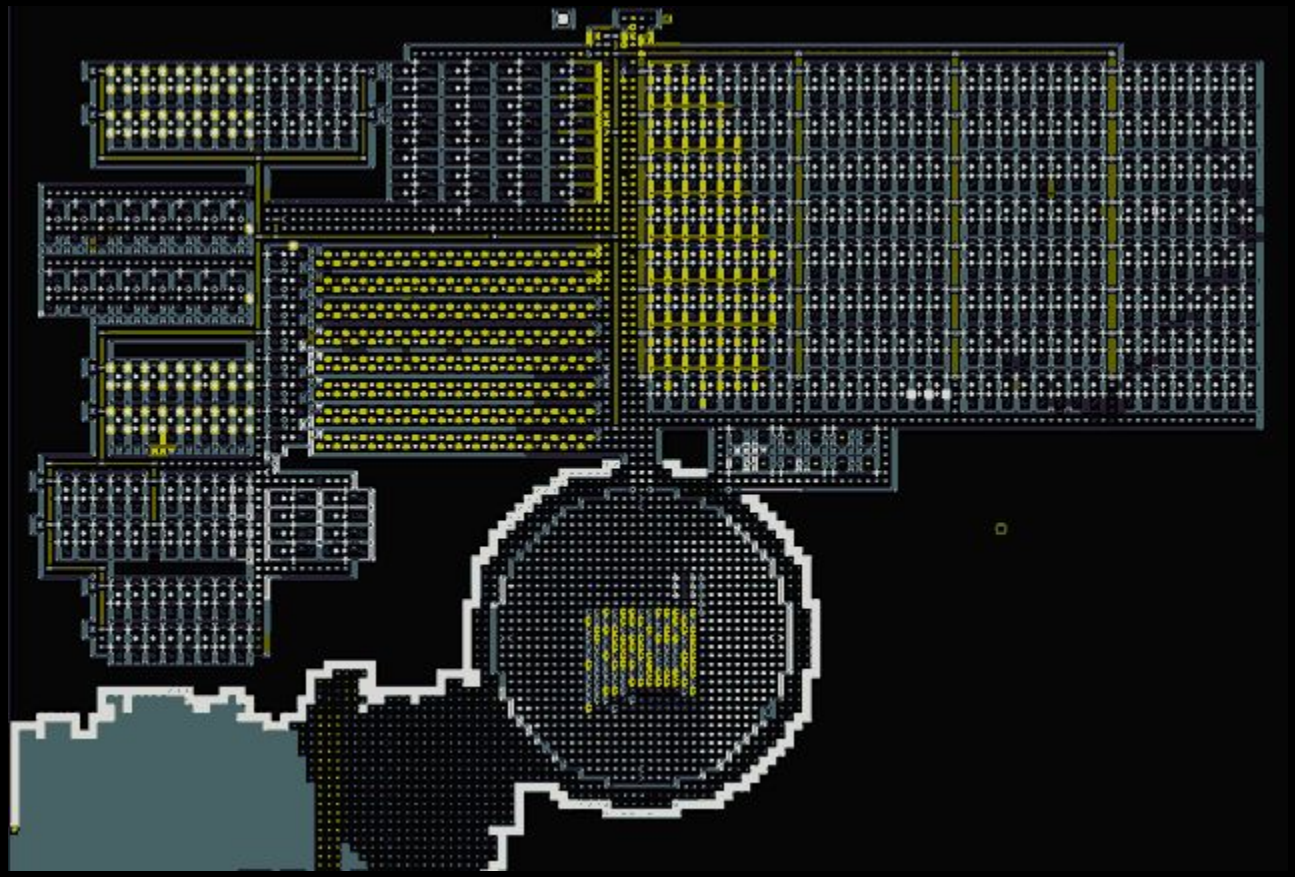
©1995, 96, 98 Nintendo, Creatures, GAMEFREAK.

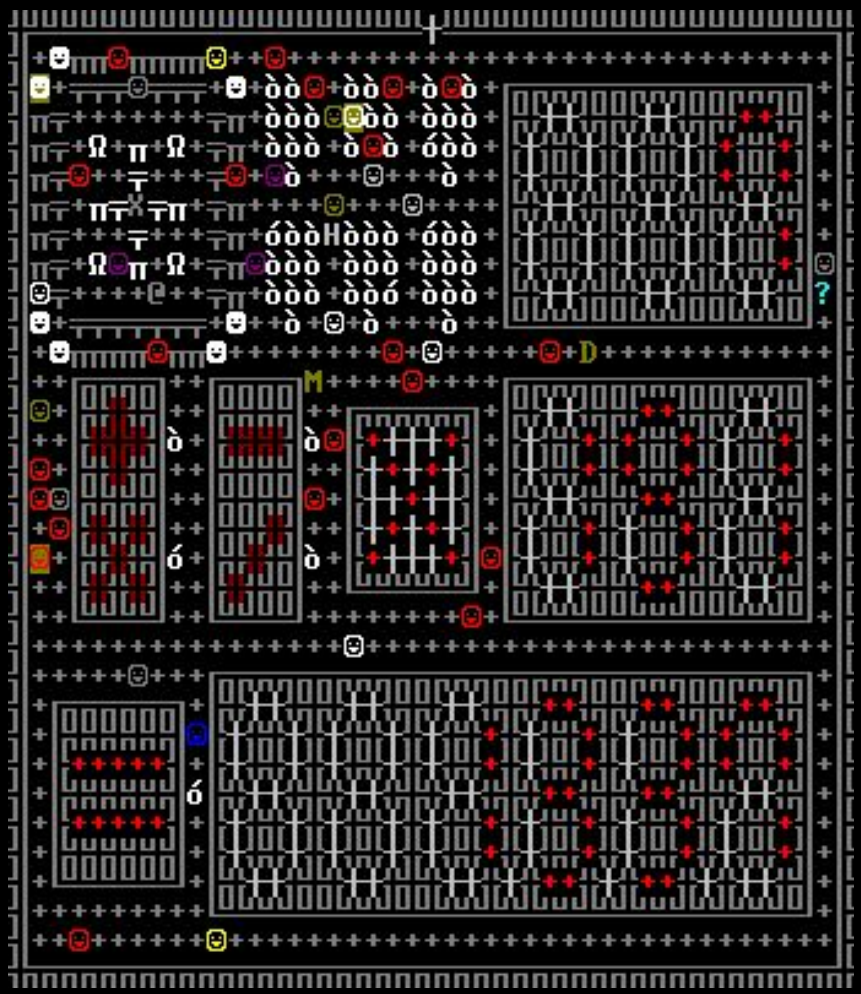

Fare i conti con i treni
Westerbrücken (11,386)
Westerbrücken East
Westerbrücken Sägewerk
Westerbrücken Sägewerk Ab
Westerbrücken Sägewerk An
Westerbrücken Hbf
Westerbrücken Station #270
Westerbrücken Gbf Vieh
Westerbrücken Gbf Getreide
Westerbrücken Gbf
Westerbrücken South
Flensberg Branch
Westerbrücken Gbf Holz
Westerbrücken Gbf Bretter
Upper Flensberg
Westerbrücken Gbf Kohle
Timetable
Go To
6th Apr 1942
** PAUSED **
30,096,084 RM


Dwarf Fortress


YO DAWG I HEARD YOU LIKE COMPUTING
SO I PUT A COMPUTER IN YOUR COMPUTER
SO YOU CAN COMPUTE WHILE YOU COMPUTE

THE DWARVEN COMPUTER







Ecco una foto del nostro pappagallo per ringraziarvi dell'attenzione.

Domande?

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